

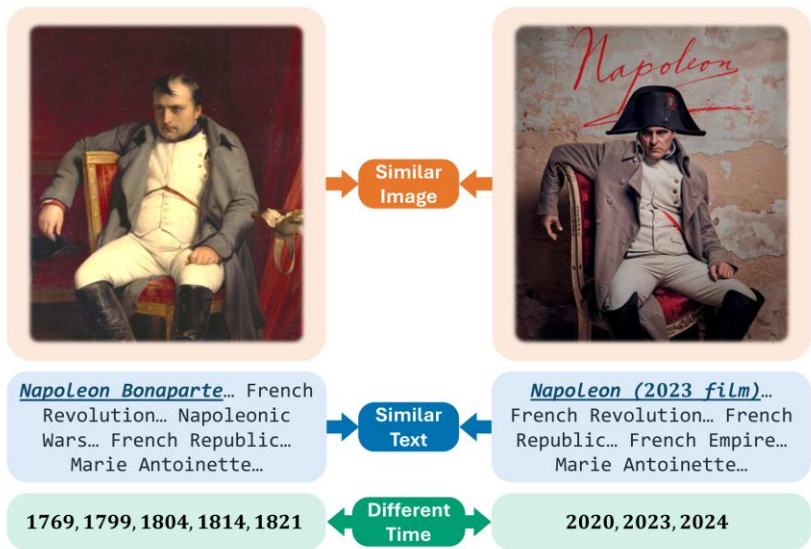
**I build models that keep telling
similar entities apart
as time changes.**

Pengyu Zhang

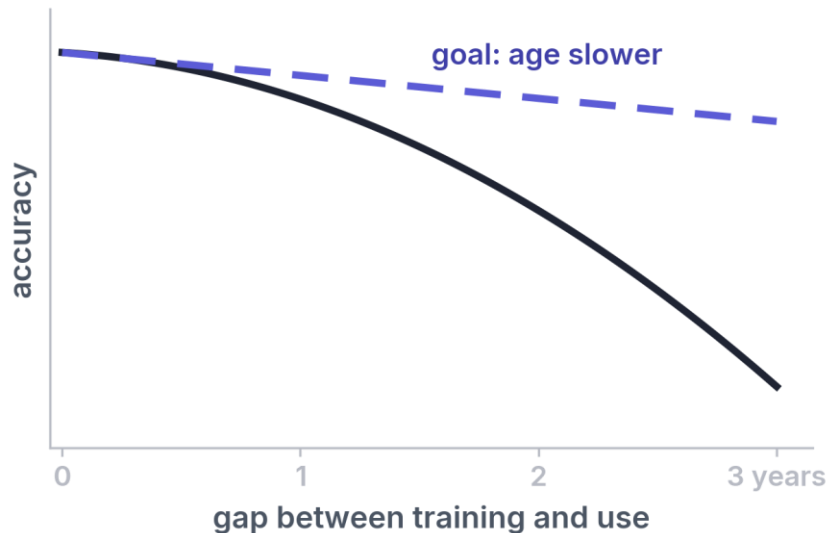
PhD candidate, University of Amsterdam · INDElab
defense expected early 2027 · pengyu-zhang.github.io

Entity ambiguity is hard on its own, temporal drift makes it harder.

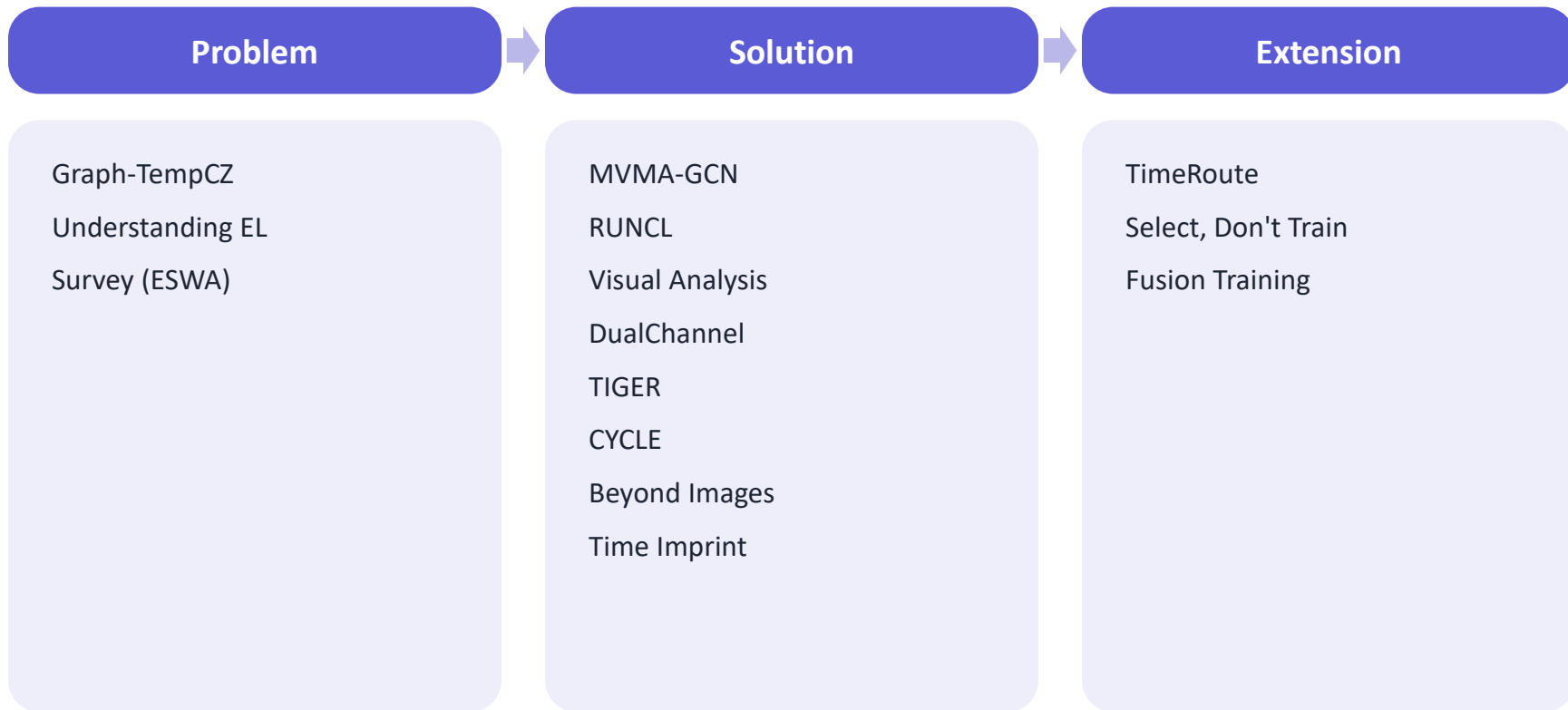
Entity ambiguity



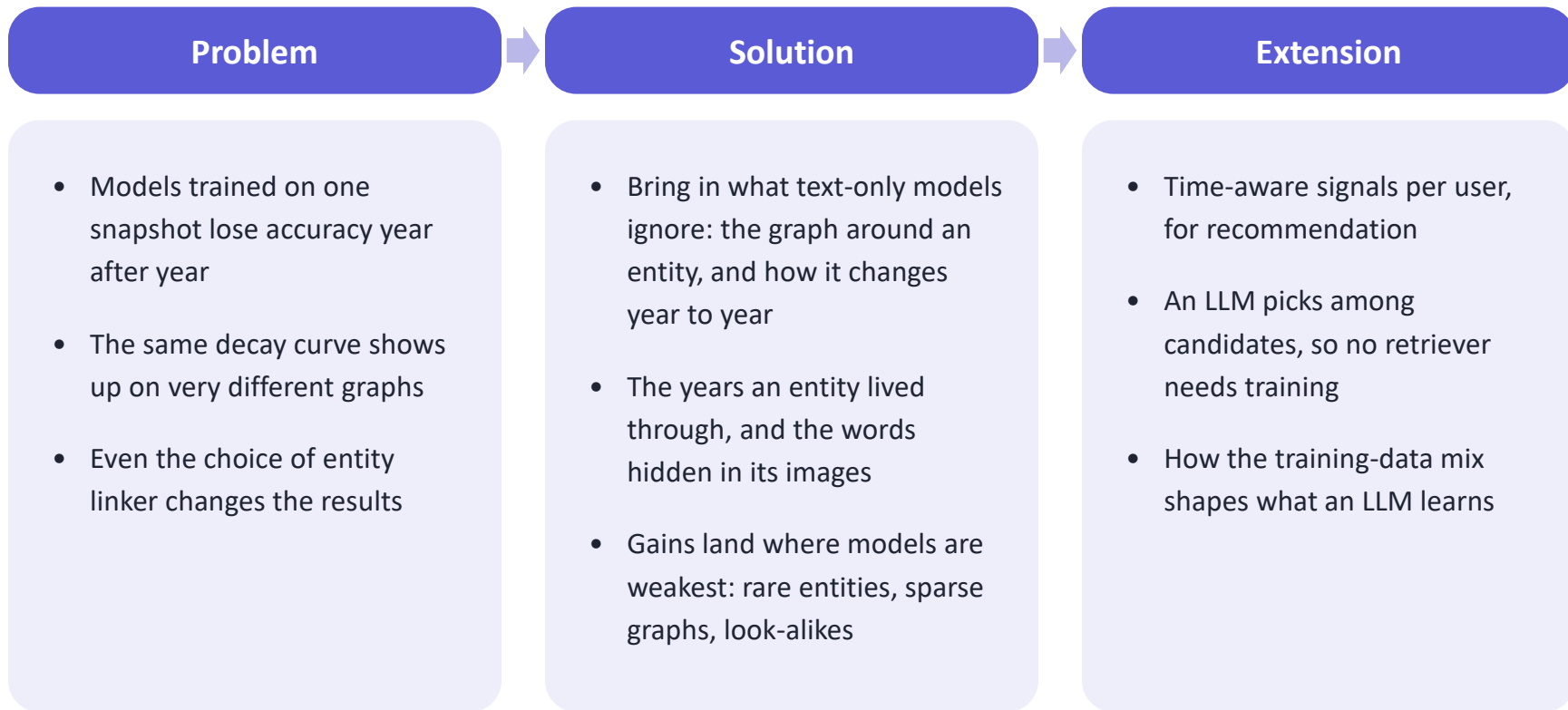
Temporal drift



Diagnose the failure, add the missing signal, reuse it elsewhere.



Diagnose the failure, add the missing signal, reuse it elsewhere.

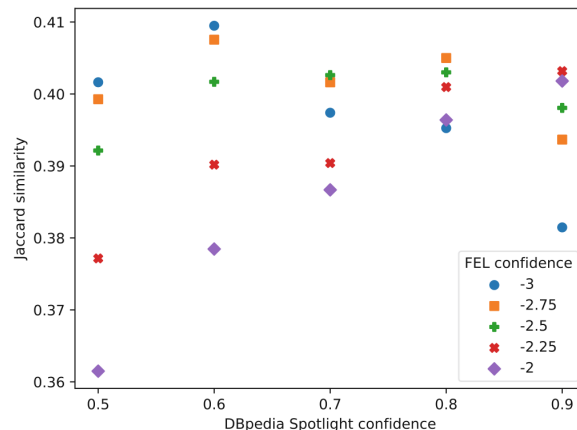


Problem: diagnose the failure

		Accuracy										
Train \ Test		2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
		2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
2010		0.8645	0.8557	0.8475	0.8367	0.8358	0.8311	0.8325	0.8295	0.8294	0.8273	0.8246
2011		0.8687	0.8656	0.8515	0.8450	0.8405	0.8370	0.8374	0.8365	0.8365	0.8334	0.8339
2012		0.8799	0.8742	0.8761	0.8661	0.8602	0.8594	0.8585	0.8583	0.8577	0.8542	0.8540
2013		0.8933	0.8925	0.8860	0.8883	0.8712	0.8737	0.8716	0.8706	0.8692	0.8673	0.8651
2014		0.8906	0.8941	0.8878	0.8896	0.8931	0.8777	0.8775	0.8765	0.8759	0.8750	0.8733
2015		0.8914	0.8913	0.8901	0.8890	0.8890	0.8965	0.8797	0.8798	0.8788	0.8793	0.8776
2016		0.8849	0.8882	0.8876	0.8900	0.8912	0.8929	0.9038	0.8845	0.8840	0.8826	0.8817
2017		0.8885	0.8869	0.8899	0.8917	0.8941	0.8944	0.8945	0.9068	0.8867	0.8865	0.8857
2018		0.8989	0.8981	0.8997	0.9016	0.9022	0.9030	0.9051	0.9076	0.9136	0.8997	0.8976
2019		0.8861	0.8877	0.8868	0.8895	0.8893	0.8917	0.8934	0.8952	0.8945	0.9109	0.8826
2020		0.8888	0.8907	0.8955	0.8960	0.8975	0.8992	0.9018	0.9045	0.9061	0.9068	0.9200

Best in the training year,
worse every year after

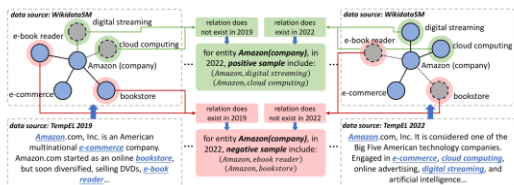
6M software mentions, one model per year, tested on every year



Two linkers agree on
only ~40% of entities

21M tweets, two entity linkers

Solution: add the missing signal



the graph and its change

≈21%

over the 2023 state of the art
at a 3-year gap, widening with time

TIGER · CYCLE

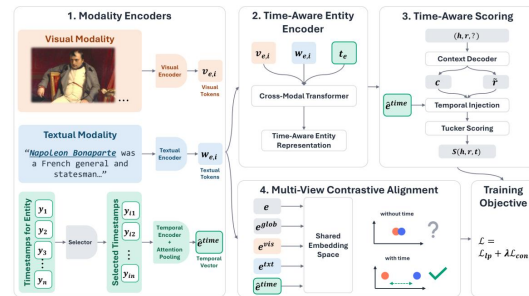


the words hidden in images

7.5 → 32.5

Hits@1 on a logo-only subset,
downstream model unchanged

Beyond Images



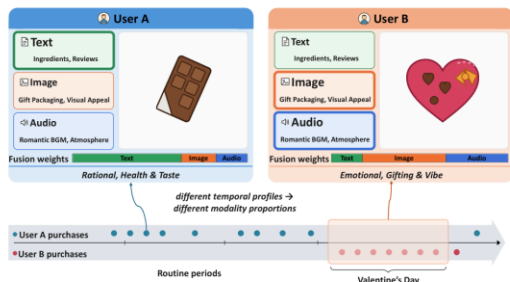
time itself

+58%

Hits@1 on the hardest 1%
of look-alike entities

Time Imprint

Extension: reuse it elsewhere

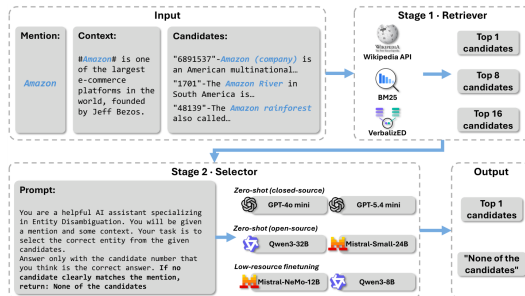


time signal, per user

5–10%

over the strongest baseline
in recommendation, ten seeds

TimeRoute

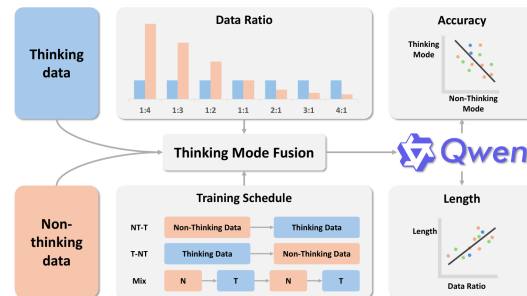


let an LLM pick

86.3 vs 82.3

micro-F1 on ZELDA,
no retriever training

Select, Don't Train · with F. Polat



what data to feed an LLM

$r = -0.58$

more short-answer data,
weaker long reasoning

Fusion Training · with C. Cao

What I've learned

1

Change is signal, not noise.

The diff between two versions of a graph is free supervision. So is when a user is active. Most systems file that under bookkeeping.

For a real system: If your catalogue changes or your logs have timestamps, the training signal is already there.

2

Time is a modality, not a feature.

When text and image cannot separate two entities, the years often can.

For a real system: In matching and deduplication, the timestamp belongs next to the text and the picture, not in a column at the end.

3

The added signal helps most where the old signal was weakest.

Rare entities, sparse graphs, unreadable logos, users with unusual behaviour. The same pattern in every project.

For a real system: The errors that matter live in the long tail. Evaluate on that subset; the average will say nothing happened.

And a small one: more than once the cheapest win was changing the data, not the model.

What's next, and where it applies

Open questions

- **When does another signal stop paying for itself?**
Every method here adds a component. The cost-benefit boundary is unmapped.
- **Continuous change, not yearly snapshots.**
Added and removed edges are treated alike today; that is probably wrong.
- **Beyond Wikidata.**
What breaks on a product catalogue or a domain knowledge graph?

Where it lands

- **Search and recommendation.**
Models drift between retrains; per-user timing changes what to trust.
- **Catalogue and entity matching.**
Look-alikes, the long tail, author disambiguation, deduplication.
- **Knowledge bases behind RAG.**
They go stale the way my entity linkers did, and could be updated selectively instead of rebuilt.

Graph-TempCZ

LREC 2026 · Congfeng Cao (lead), Pengyu Zhang, Jelke Bloem

Accuracy											
Test \ Train	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
2010	0.8645	0.8557	0.8475	0.8367	0.8358	0.8311	0.8325	0.8295	0.8294	0.8273	0.8246
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2020	0.8888	0.8907	0.8955	0.8960	0.8975	0.8992	0.9018	0.9045	0.9061	0.9068	0.9200

figure from the paper

93%

accuracy for GraphSAGE, about 6 points above the best feature-based baseline

Problem Software-usage data had no graph structure and no time axis.

What we did Built a publication–software graph from 6.5M mentions in biomedical papers and framed usage prediction as temporal link prediction; one model per year, 2010–2020, tested on every year.

Result Best in the training year, decaying with the gap; about a two-year transfer window. We; I helped with the paper logic, figures and revisions.

Understanding EL

EKAU 2024 · James Nevin (lead), Pengyu Zhang, D. Dimitrov, M. Lees, P. Groth, S. Dietze

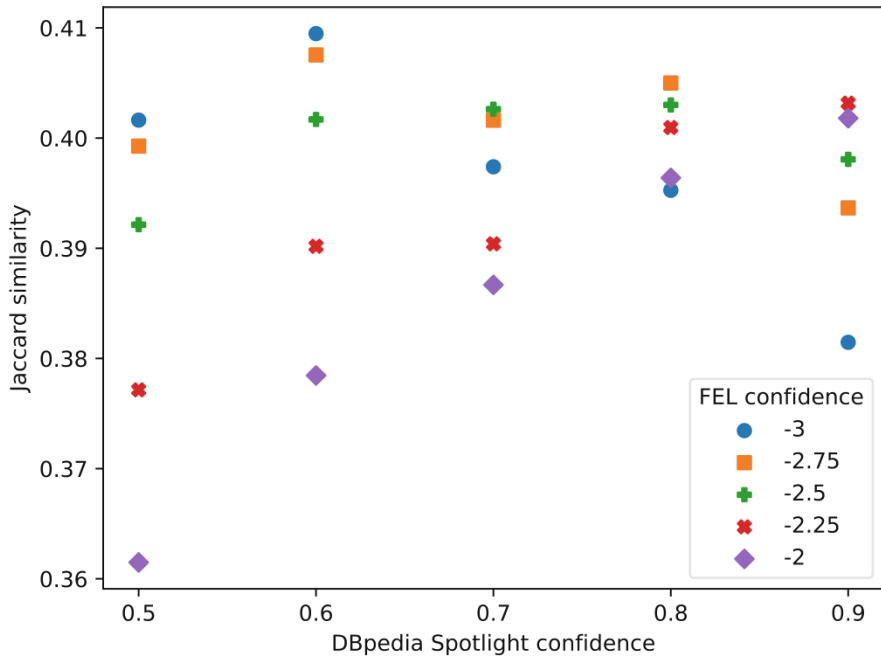


figure from the paper

~40%

overlap between the entities two linkers extract from the same 21M tweets (Jaccard 0.38–0.41)

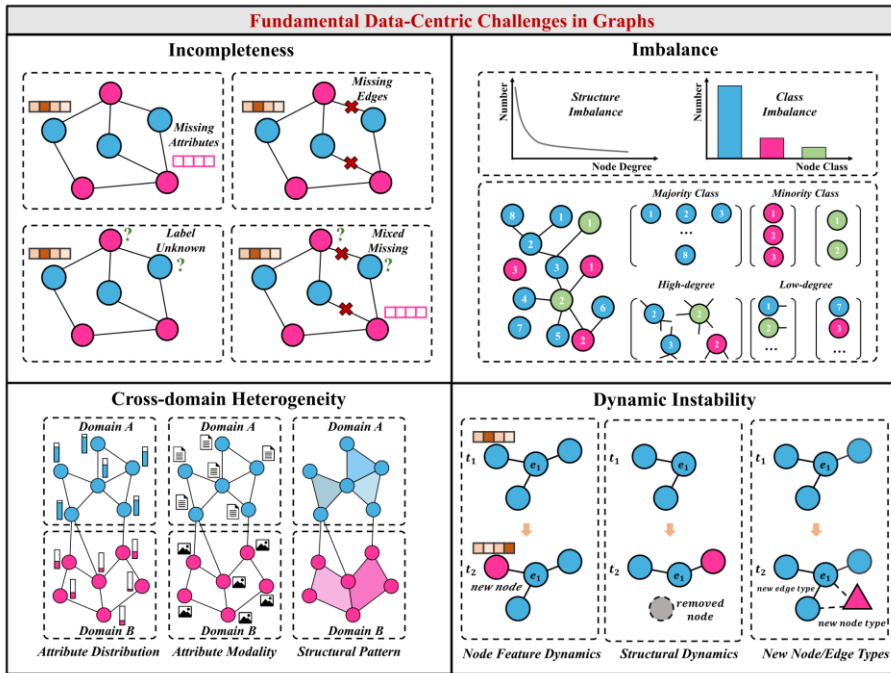
Problem Tweets are short and noisy; without ground truth you cannot pick the best linker, yet the downstream network depends entirely on it.

What we did Linked 21M COVID-era tweets with FEL and DBpedia Spotlight at five thresholds each and compared the resulting co-occurrence networks. I owned the entity linking component.

Result Network size, connectivity and top entities all shift with tool and threshold; COVID-19 drops out of the top ten at strict FEL settings. Report across linkers, not one.

Survey: LLMs for data challenges in graphs

ESWA 2026 · Mengran Li (lead), Pengyu Zhang, and 12 co-authors (UvA, SYSU, BJUT)



380+

papers kept from more than a thousand screened,
organised by four data challenges

Problem Real graph data is incomplete, imbalanced, heterogeneous across domains and keeps changing; LLM work on graphs lacked a data-centric map.

What we did Organised the literature by the four data challenges rather than by model type, each with traditional and LLM approaches; open repository. I wrote the dynamic instability and temporal sections.

Result A taxonomy and reading map; open problems in efficiency, interpretability, continual adaptation and hallucination.

figure from the paper

MVMA-GCN

EAAI 2023 · Pengyu Zhang, Yong Zhang, Jingcheng Wang, Baocai Yin (BJUT)

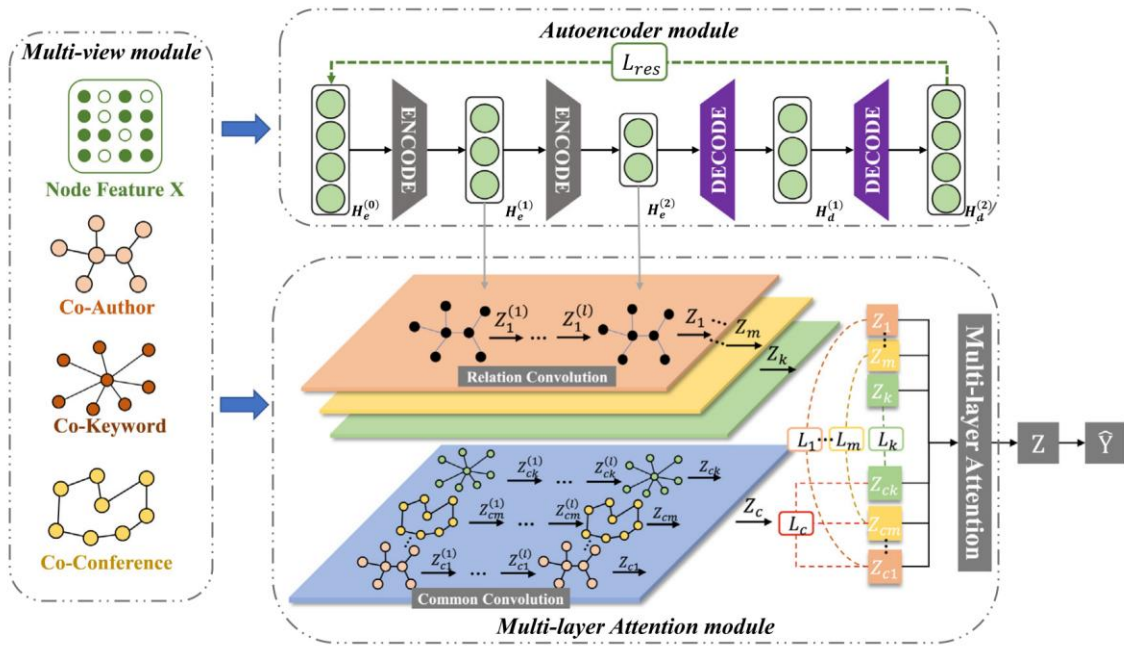


figure from the paper

+2 pts

over AM-GCN on ACM (92.7 vs 90.4 accuracy at 20 labels per class); best on most of eight benchmarks

Problem Real networks have several kinds of links; single-view GNNs throw that away and cannot weight views or neighbours adaptively.

What I did All relation views plus a kNN feature graph convolved together; HSIC keeps each view's representation distinct; one attention layer over neighbours, one over views; an autoencoder branch keeps feature structure.

Result Best in most settings across eight datasets; each module contributes in ablation. The distinct/shared idea and HSIC carried straight into TIGER.

RUNCL

Physica A 2024 · Pengyu Zhang, Yong Zhang, Xinglin Piao, Yongliang Sun, Baocai Yin (BJUT)

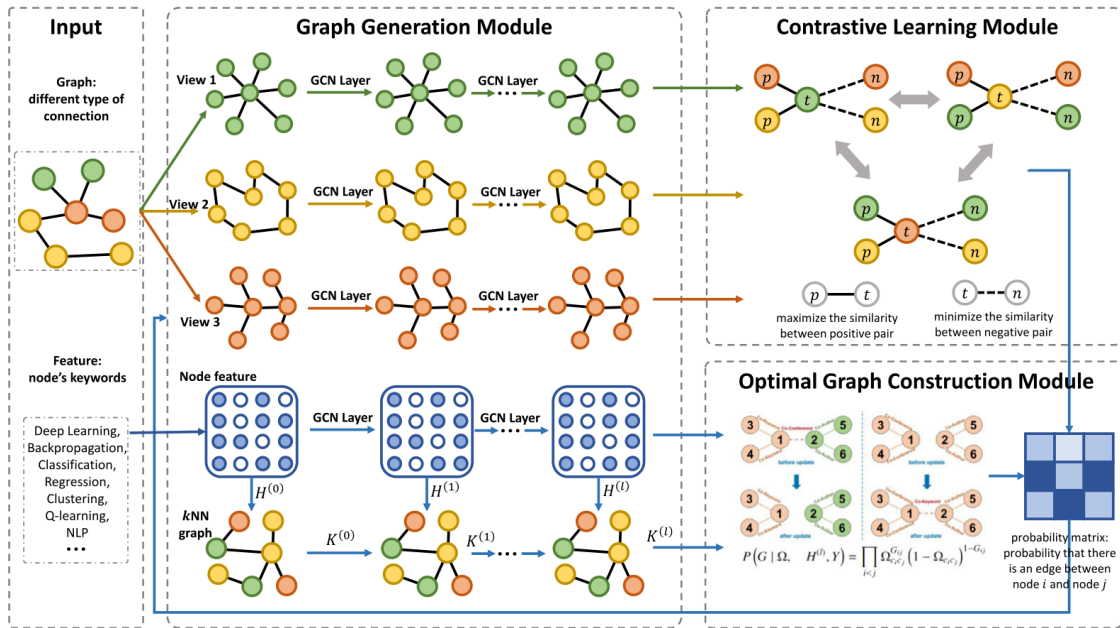


figure from the paper

3% → 6%

gain over the strongest baseline on low-degree nodes as labels shrink from 20 to 5 per class

Problem Missing or wrong edges hurt poorly connected nodes most, and GNNs treat the input graph as ground truth.

What I did Generate candidate kNN neighbourhoods from node features, pick the best graph with a stochastic block model and Bayesian inference, then contrastive learning between the feature view and the relation view.

Result Best in most low-label settings on six benchmarks; the contrastive part matters most when labels are scarcest. The low-degree focus and cross-view contrastive carried into CYCLE.

Visual Analysis for name disambiguation

JCAD 2022 (in Chinese) · Pengyu Zhang, Yong Zhang, Yanjie Cui, Baocai Yin (BJUT)



95%+

task success for eleven volunteers on 4,000 real university papers; usability 84–88 / 100

Problem Romanised Chinese names collide; hundreds of authors under one spelling, and existing tools were hard for non-experts to use or verify.

What I did A multi-view GCN classifier (co-author, co-venue, co-keyword, kNN features) predicting each author's field, plus linked views: correlation, collaboration network with journal linkage, raw metadata.

Result Classifier about 91% on ACM and DBLP; two real cases confirmed by research administrators; user study with students and administrators.

figure from the paper

DualChannel

Information 2021 · Xin Zheng (lead), **Pengyu Zhang**, Yanjie Cui, Rong Du, Yong Zhang (BJUT)

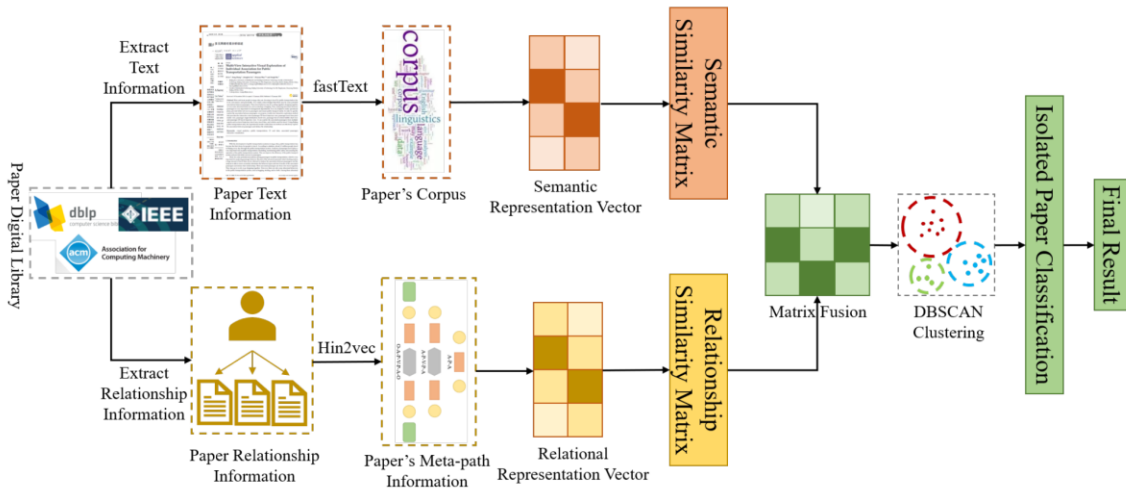


figure from the paper

F1 0.62

on AMiner WholsWho, above six baselines, with balanced precision and recall

Problem Name disambiguation methods used either text or graph structure, rarely both.

What we did A semantic channel (fastText on an academic corpus) and a relation channel (co-author, organisation, venue heterogeneous graph with meta-path walks, Hin2Vec); similarity matrices added, DBSCAN clustering.

Result F1 0.62 versus 0.61 for the best baseline; works best for names with fewer homonyms and enough history. We; I wrote the manuscript with all figures and tables.

TIGER

ECAI 2024 · Pengyu Zhang, Congfeng Cao, Paul Groth

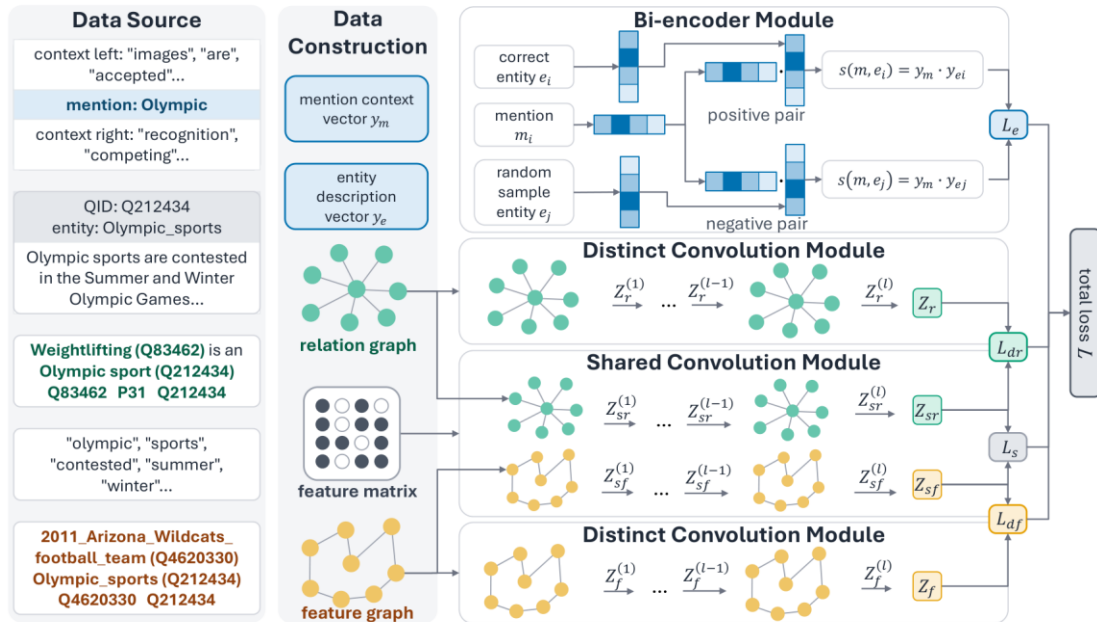


figure from the paper

+21%

over SpEL on continual entities at a 3-year gap (16% at one year)

Problem Text-only linkers decay as the knowledge graph evolves.

What I did Add Wikidata relations and a description-similarity graph; shared and distinct GCN modules with an HSIC loss keep what each graph knows. Graph at training, text-only at inference.

Result Gains grow with the gap; largest on well-connected entities. Built Graph-TempEL, four yearly snapshots, 136k entities.

CYCLE

CIKM 2024 · Pengyu Zhang, Congfeng Cao, Klim Zaporozhets, Paul Groth

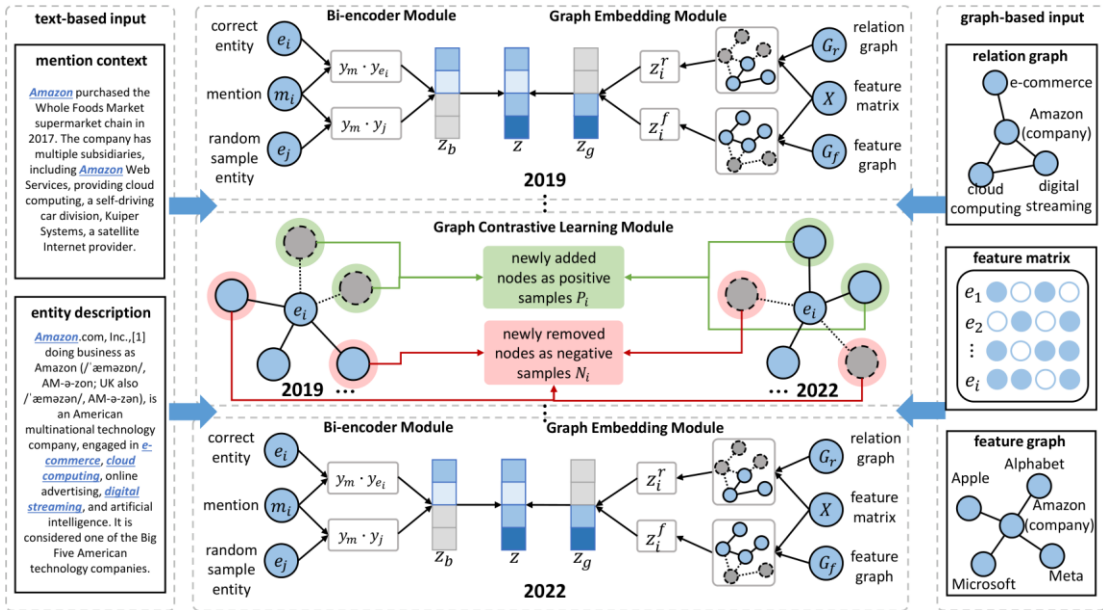


figure from the paper

+18%

over SpEL at a 3-year gap (14% at one year), continual and new entities combined

Problem TIGER treats each year's graph as static; the change itself was unused.

What I did Edges that appear between snapshots become positives, edges that disappear become negatives; cross-view contrastive learning between relation and feature graphs.

Result Largest gains on low-degree entities, the ones most exposed to drift. Built GCL-TempEL.

Beyond Images

ESWC 2026 · Pengyu Zhang, Klim Zaporozets, Jie Liu, Jia-Hong Huang, Paul Groth

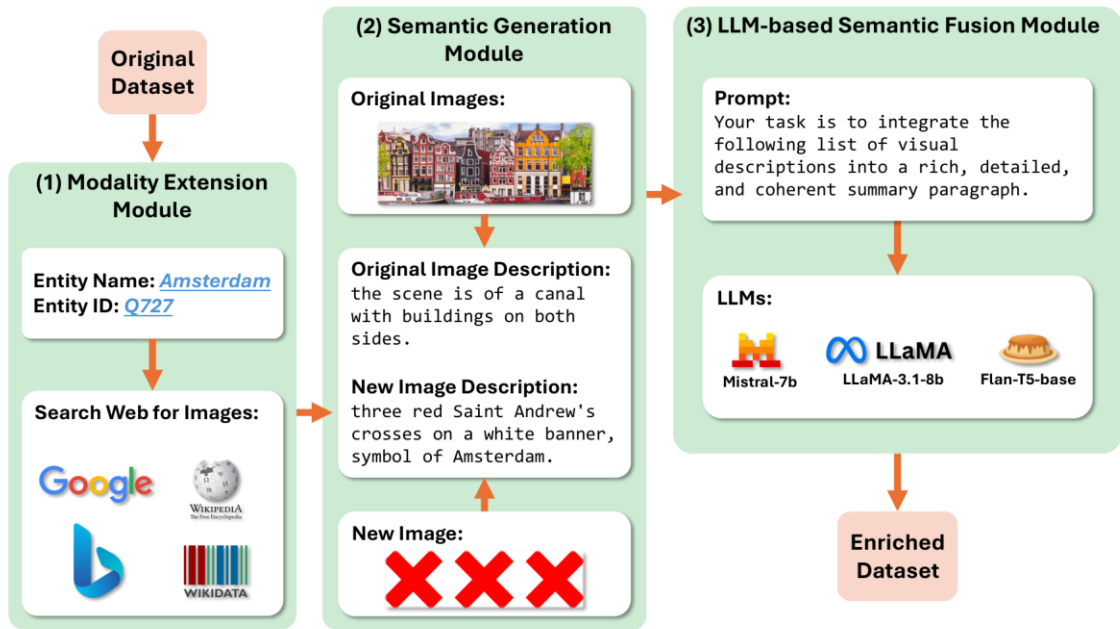


figure from the paper

7.5 → 32.5

Hits@1 on a logo-and-symbol subset; +5 to 8% overall across four models

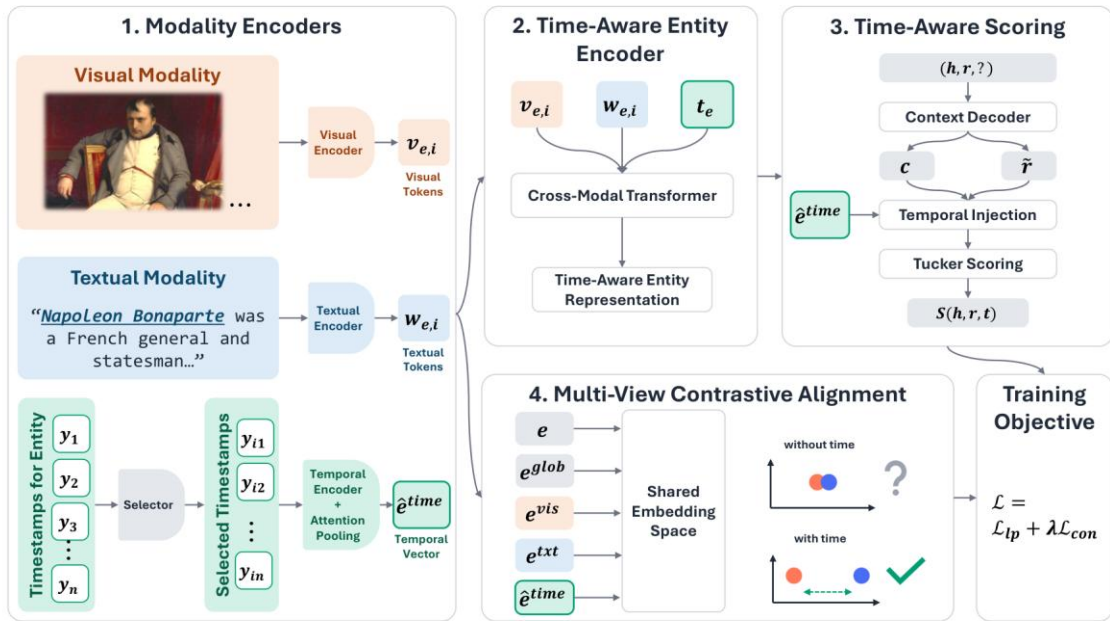
Problem Curated multi-modal KGs drop ambiguous images because their visual embeddings are noise.

What I did Retrieve every Wikipedia image for the entity, caption each with a VLM, fuse the captions with an LLM into one summary, use it as the text input.

Result No downstream model changes; all four models improve on three datasets. Enrichment runs once, offline.

Time Imprint

under review · Pengyu Zhang, Klim Zaporozhets, Congfeng Cao, Jia-Hong Huang, Paul Groth



+58%

Hits@1 on the top-1% most ambiguous entities; +6%
Hits@1 overall on MKG-W

Problem Look-alike entities share text and images; time lives at triple level, not entity level.

What I did Select a few timestamps around the median plus the earliest year, pool with attention, inject as prefix token, into scoring with a relation gate, and as a contrastive view.

Result New state of the art on MKG-W. Still within one point at 40% timestamp corruption.

figure from the paper

TimeRoute

under review · Pengyu Zhang, Yangqin Jiang, Klim Zaporozhets, Congfeng Cao, Paul Groth

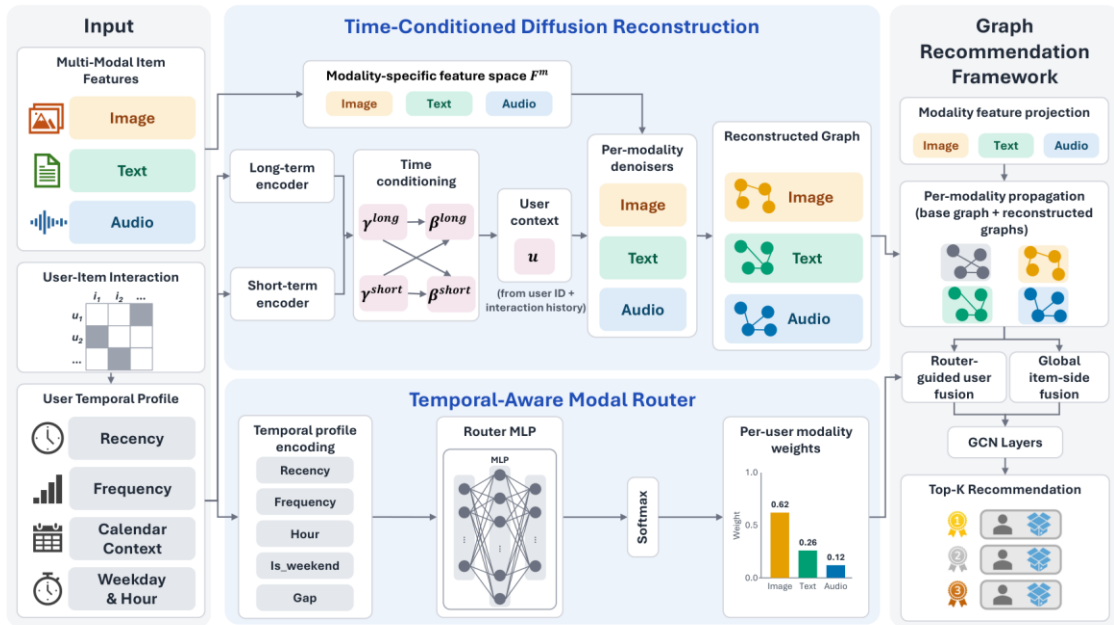


figure from the paper

5–10%

over the strongest baseline on TikTok and two Amazon datasets, ten seeds

Problem Modality relevance drifts per user; one global fusion weight cannot track it.

What I did A router turns each user's timestamp profile into modality weights (under 700 parameters); a time-conditioned diffusion module filters outdated edges before graph propagation.

Result Random noise into the router removes the gain, so it is the temporal signal. Routing shifts with user activity.

Select, Don't Train

ISWC 2026 · Fina Polat (lead), Daniel Daza, Pengyu Zhang, Klim Zaporjets, Paul Groth

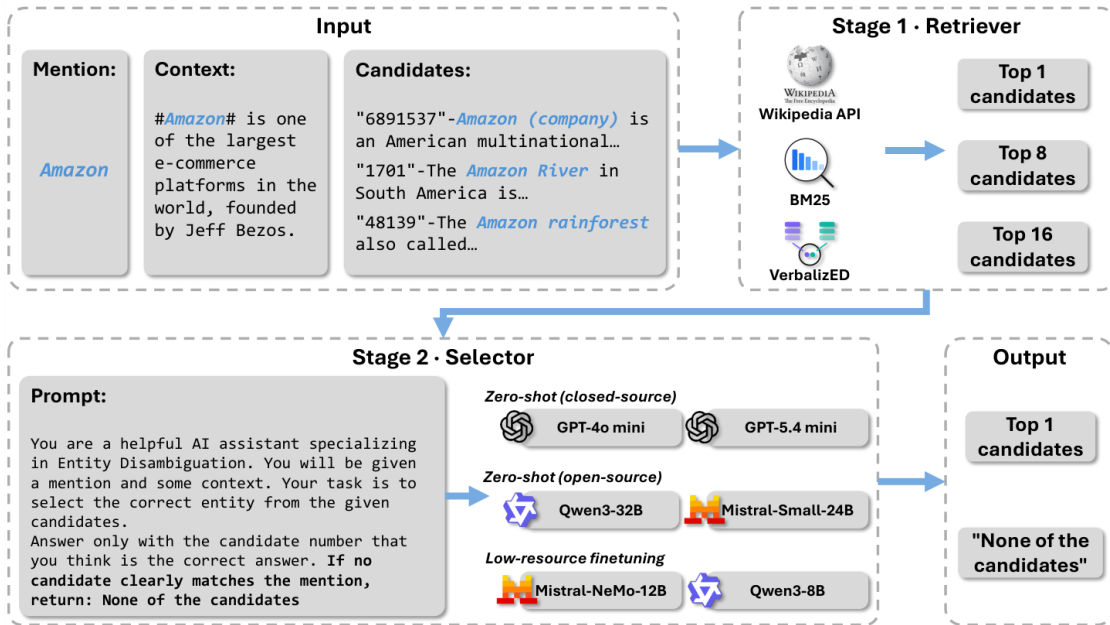


figure from the paper

86.3

micro-F1 on ZELDA with untrained BM25 + GPT-5.4-mini, 4 points above the best dual encoder

Problem Dual encoders squeeze retrieval and selection into one space and need retraining as the KB changes.

What we did Split them: a retriever returns 16 candidates, an LLM selects one or abstains. Retrievers: Wikipedia API, BM25, or a trained dense model.

Result A trained dense retriever lifts it to 88.5; rewarding correct abstention, 90.7. We; Fina led, I was in the discussions throughout.

Fusion Training

ACL 2026 SRW · Congfeng Cao (lead), Pengyu Zhang, Jelke Bloem

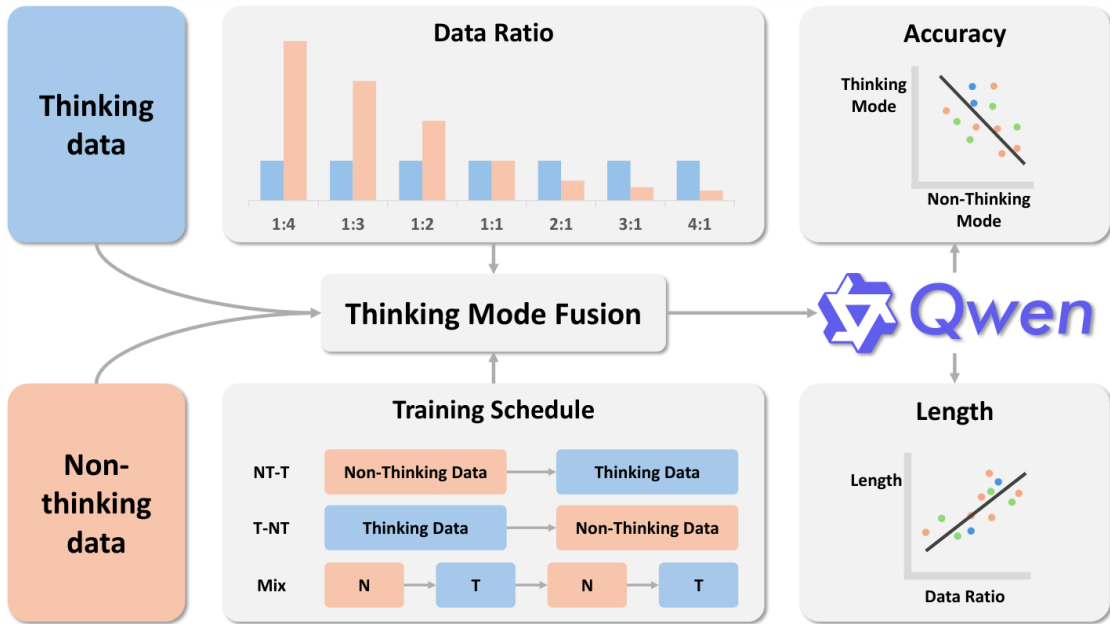


figure from the paper

$$r = -0.58$$

correlation between thinking and non-thinking accuracy across ratios and schedules; interleaved training most robust

Problem Hybrid reasoning models like Qwen3 are fine-tuned on both long-reasoning and short-answer data; the effect of the ratio and the order was unstudied.

What we did Fusion Bench: thinking data fixed at 1,500, non-thinking from 375 to 6,000 (seven ratios), three schedules (T-NT, NT-T, interleaved), Qwen3-4B with LoRA, three runs each.

Result More short-answer data lowers long-reasoning accuracy (0.224 $\rightarrow$ 0.199); the best schedule depends on the ratio; interleaving is the most robust. We; I helped with the paper logic, figures and revisions.

How the thesis puts it

Addressing Temporal Dynamics and Entity Ambiguity in Knowledge Graph Tasks with Multi-Source Information · SIKS 2026

Hypothesis: information sources not yet used as modeling signals improve KG tasks under temporal dynamics and entity ambiguity				
Challenge 1: Temporal dynamics (RQ1)		Challenge 2: Entity ambiguity (RQ2)		
Task	Chapter 2 · RQ1.1 TIGER + CYCLE	Chapter 3 · RQ1.2 TimeRoute	Chapter 4 · RQ2.1 Beyond Images	Chapter 5 · RQ2.2 Time Imprint
	Entity Linking	Multi-modal recommendation	Link prediction in MMKGs	Link prediction in MMKGs
	Text-only linkers degrade as the KG evolves	Modality relevance drifts per user; global fusion cannot track it	Ambiguous images (logos, symbols) act as noise and are discarded	Look-alike entities share text and images
	Graph structure and its evolution	Users' temporal contexts	Text distilled from ambiguous images	Time as an entity-level modality
	Graph convolution over snapshots + cross-year edge contrastive learning	Per-user modal router + time-conditioned diffusion reconstruction	Image retrieval + image-to-text conversion + LLM fusion	Three-view contrastive alignment + timestamp selection
Finding	Complementary gains across high- and low-degree entities	Gains need both the mechanisms and temporal input	Largest gains on logo and symbol images	Largest gains on the most ambiguous entities